In the previous exercise, we note that the result was based on knowledge of the distribution of the test statistic under the null hypothesis. It is key that the test statistic be chosen so that its distribution is known (at least approximately), under the null. As usual, the central limit theorem can be very helpful.

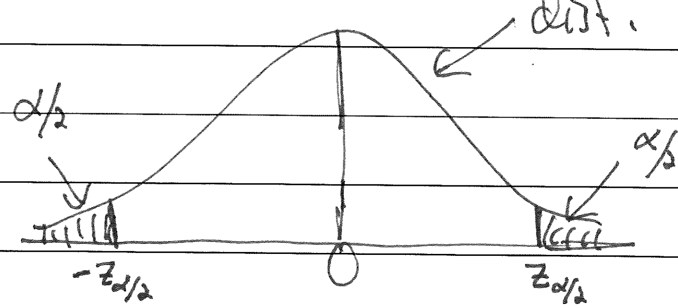
Also, we note that in the exercise we tested the one-sided alternative hypothesis $H_1 : \mu > 10$. In other cases we will test a two-sided alternative, for example, $H_1 : \mu \neq 10$. In such a case the null would be rejected for either small or large values of the test statistic.

Exercise: Suppose that we tested $H_1 : \mu \neq 10$ in the above example. What would the rejection region be for fixed α?

$(-\infty, c_1) \cup (c_2, \infty)$ with $T = \bar{X}$

$c_1 < 10 < c_2$

dist. of $\dfrac{\bar{X}-10}{\sigma/\sqrt{n}}$ under H_0



Reject if $\dfrac{\bar{X}-10}{\sigma/\sqrt{n}} > z_{\alpha/2}$ or $\dfrac{\bar{X}-10}{\sigma/\sqrt{n}} < -z_{\alpha/2}$

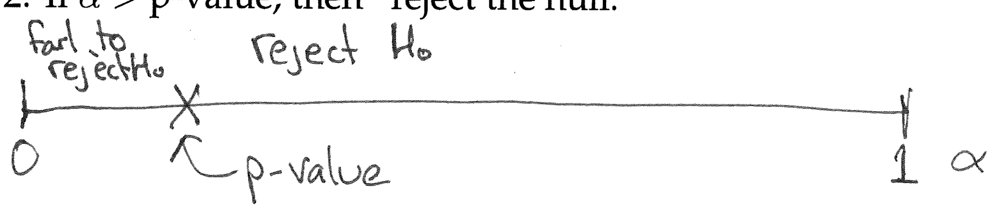
The p-value

The choice of α in the statistical hypothesis testing framework may seem arbitrary. Why would one choose one value of α over another?

As α is chosen smaller, the experimenter is choosing to be more cautious, to protect against Type I error. The opposite is true as α is chosen larger.

For a given data set, there is a **dividing line** between the choices of α for which the conclusion is "reject the null" and the conclusion is "fail to reject the null." This dividing line is the p-value.

1. If $\alpha <$ p-value, then "fail to reject the null."
2. If $\alpha >$ p-value, then "reject the null."



It is important to note that the p-value is **a function of the data**.

The p-value is **not** the probability that the null hypothesis is true. Instead, it is the largest choice of α such that the conclusion is "fail to reject the null."

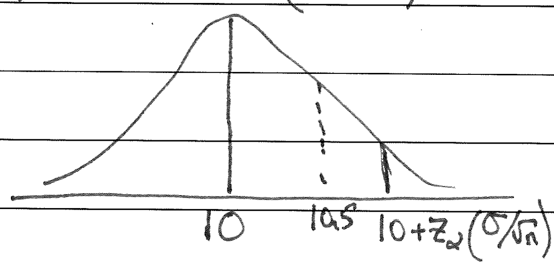
The appeal is that one can report a p-value, and allow the reader to decide. Most software packages report the results of hypothesis tests as p-values.

The next exercise illustrates another interpretation of the p-value: It is the probability that, assuming the null hypothesis is true, the test statistic would be equal to or more extreme in favor of the alternative than the observed test statistic.

This interpretation is often useful for making computations.

Exercise: Return to our example when testing $H_0 : \mu = 10$ versus $H_1 : \mu > 10$. Assume that $\sigma^2 = 1$. We observe $\overline{x} = 10.5$ with a sample size of $n = 50$. What is the p-value in this case? What happens if we switch to $H_1 : \mu \neq 10$?

When testing $H_0: \mu = 10$ vs. $H_1: \mu > 10$, we reject if $\bar{x} > 10 + z_\alpha (\sigma/\sqrt{n})$



When would we reject?

If α is chosen so that

$10 + z_\alpha (\sigma/\sqrt{n}) < 10.5$, then reject H_0

If α is chosen so that

$10 + z_\alpha (\sigma/\sqrt{n}) > 10.5$, then fail to reject H_0

What choice of α puts us on the dividing line?

$10 + z_\alpha (\sigma/\sqrt{n}) = 10.5$

i.e. $z_\alpha = (0.5)\left(\frac{\sqrt{n}}{\sigma}\right) = (0.5)\frac{\sqrt{50}}{1} = 3.53$

so, $\alpha = P(Z > 3.53) = 0.000208$

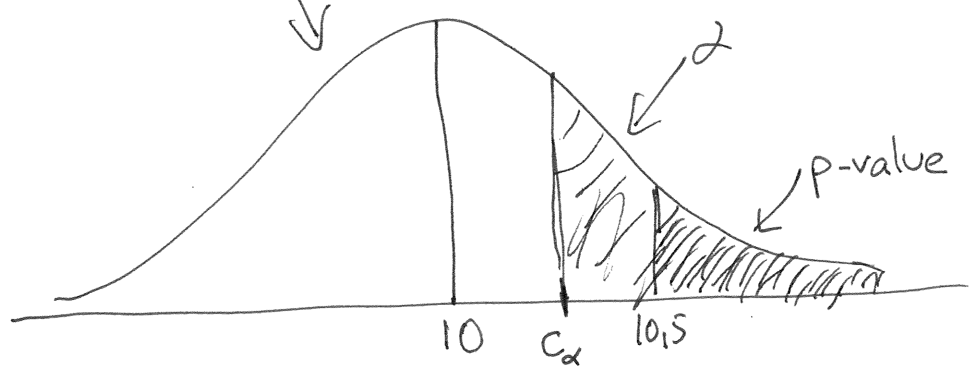
That is the p-value.

"Only people who choose $\alpha < 0.000208$ would fail to reject H_0 in this case."

Another way of thinking of it:

What choice of α forces the cutoff to be 10.5?

It's exactly the prob. above 10.5 when using the $N(10, \sigma^2/n)$ dist.



$c_\alpha = 10 + z_\alpha(\sigma/\sqrt{n})$

$P(X > c_\alpha) = \alpha, \ X \sim N(10, \sigma^2/n)$

The general result:

p-value = P(test stat is equal to the obs. test stat or is more extreme in the direction of the alternative | H_0 is true)

In the two-sided case: $H_0: \mu = 10$ vs. $H_1: \mu \neq 10$, $\bar{X} = 10.5$

$$\text{p-value} = P(\bar{X} \geq 10.5 \text{ or } \bar{X} \leq 9.5 \mid H_0 \text{ is true})$$

$$= 2P(\bar{X} \geq 10.5 \mid H_0 \text{ is true})$$

The Wald Test

The Wald Test is built around a simple notion: An ideal choice for the test statistic is the number of standard errors a parameter estimate is from the null hypothesized value for that parameter.

Specifically, if testing $H_0\colon \theta = \theta_0$, then the test statistic is chosen as

$$T = \frac{\widehat{\theta} - \theta_0}{\mathrm{SE}(\widehat{\theta})}$$

T gives the "the number of standard errors $\widehat{\theta}$ is from θ_0."

In a wide range of situations, T has approximately the standard normal distribution under the null hypothesis. Hence, the region region is not difficult to derive.

Exercise: What is one very important case in which we can state that T is approximately standard normal under the null?

$\hat{\theta}$ is the MLE for θ

$$SE(\hat{\theta}) = 1 / \sqrt{n I(\theta_0)}$$

Since $\hat{\theta}$ is approx. normal,

$$T = \frac{\hat{\theta} - \theta_0}{\sqrt{1 / n I(\theta_0)}}$$ is approx. standard normal under $H_0 : \theta = \theta_0$

The three possible tests can be presented as follows:

1. If testing $H_0\colon \theta = \theta_0$ versus $H_1\colon \theta > \theta_0$, then reject the null if $T > z_\alpha$.
2. If testing $H_0\colon \theta = \theta_0$ versus $H_1\colon \theta < \theta_0$, then reject the null if $T < -z_\alpha$.
3. If testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$, then reject the null if either $T < -z_{\alpha/2}$ or $T > z_{\alpha/2}$.

Exercise: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. Bernoulli(p). Construct the two-sided Wald Test for p. Suppose that $n = 100$ and $\sum X_i = 47$. Test $H_0\colon p = 0.4$ versus $H_1\colon p \neq 0.4$. What is the p-value?

The MLE is $\hat{p} = \sum X_i / n$

$I(p) = 1 / p(1-p)$, $SE(\hat{p}) = \sqrt{\dfrac{p(1-p)}{n}}$

So, the Wald test stat is

$$T = \frac{\hat{p} - p_0}{\sqrt{\dfrac{p_0(1-p_0)}{n}}}$$

reject H_0 if $T > z_{\alpha/2}$ or $T < -z_{\alpha/2}$

We are testing $H_0: p = 0.4$ vs. $H_1: p \neq 0.4$

Observe $\sum X_i = 47$, $n = 100$, $\hat{p} = 0.47$, $p_0 = 0.4$

$$T = \frac{0.47 - 0.40}{\sqrt{\frac{(0.4)(0.6)}{100}}} = 1.43$$

If chose $\alpha = 0.05$, $z_{\alpha/2} = 1.96$, so fail to reject H_0 when $\alpha = 0.05$

If chose $\alpha = 0.10$, $z_{\alpha/2} = 1.645$, so fail to reject H_0 when $\alpha = 0.10$

So, p-value must be larger than 0.10

What is the p-value exactly?

- Could find α such that $z_{\alpha/2} = 1.43$
- Or: p-value $= P(T \geq 1.43 \text{ or } T \leq -1.43 \mid H_0 \text{ true})$

$$= P(Z \geq 1.43 \text{ or } Z \leq -1.43)$$

$$= 2P(Z \geq 1.43) \approx 0.15$$

Choices of α larger than 0.15 would lead to rejecting H_0

Note: If testing $H_0: p = 0.4$ vs. $H_1: p > 0.4$, then p-value is $0.15/2 = P(Z \geq 1.43)$